Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	*(a)*		The specific heat capacity is the heat required to increase the temperature of 1 kg [or unit mass] of a material by 1 °C (1 degree K – or one degree/ unit temperature rise). [Alternative: equation with all terms fully defined.]	1			1		
	(b)	(i)	Idea: heat lost by boiling water=heat gained by other water (1) $1.6\left[\times 10^{-3}\right]\left(100.0-\theta_f\right)=0.6\left[\times 10^{-3}\right]\left(\theta_f-19.5\right)$ (1) θ_f = 78.0 °C (1) **Alternative (for second mark)** If $\Delta\theta$ is temperature rise of water in flask $1.6\left[\times 10^{-3}\right]\left(80.5-\Delta\theta\right)=0.6\left[\times 10^{-3}\right]\Delta\theta$ (✓)			3	3	2	
		(ii)	Mass of water = $1.6\times 10^{-3}\times 1000=1.6$ [kg] (1) Application of $m=\rho V$ [even if 0.6×10^{-3} m^3 used → 0.6 kg] [Can be credited from *(b)*(i) Heat lost = 1.6(100.0 – 78.0 **ecf**) × 4200 (1) = 1.48×10^5 J (1) 148 J → 1 mark		3		3	3	
		(iii)	[Work done is negligible as] negligible / no change in volume.			1	1		
			Question 3 total	**1**	**3**	**4**	**8**	**5**	**0**